Practical No. 7

Mapped CO: CO4 Units V

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1. Aim and Objectives

Aim: To understand Hyperledger Fabric architecture and distinguish between permissioned and permissionless blockchain technologies.

2. Theory

2.1 Blockchain Types

PERMISSIONLESS BLOCKCHAINS (Public):
• No access control - anyone can participate
• Complete decentralization
• Examples: Bitcoin, Ethereum
• Consensus: Proof of Work (PoW), Proof of Stake (PoS)
• Ledger visibility: Completely public
• Use: Cryptocurrencies, public transactions

PERMISSIONED BLOCKCHAINS (Private):
• Access control - only authorized members
• Controlled decentralization
• Examples: Hyperledger Fabric, Corda, Quorum
• Consensus: Byzantine Fault Tolerance (BFT)
• Ledger visibility: Private channels between organizations
• Use: Enterprise B2B, regulated industries

HYPERLEDGER FABRIC CHARACTERISTICS:
✓ Permissioned (certificate-based access)
✓ Modular architecture (pluggable consensus)
✓ Private channels (organization-specific data)
✓ High performance (100+ transactions per second)
✓ Instant finality (no forks)
✓ Flexible chaincode (business logic rules)

2.2 Blockchain Architecture

CORE COMPONENTS:

BLOCK STRUCTURE:
A block contains header information and transaction data. Each block is linked to the previous block through a cryptographic hash, creating an immutable chain.

• Header: Version, previous block hash, Merkle root, timestamp, difficulty
• Body: List of transactions
• Property: Changing one block invalidates all subsequent blocks

LEDGER:
Distributed database with two components:
1. Blockchain - Transaction history (immutable, append-only)
2. World State - Current values of assets (changes tracked)

Every peer maintains identical copy through consensus.

TRANSACTION:
Input (sender information) → Execution (validation) → Output (state change) → Signature (authorization)

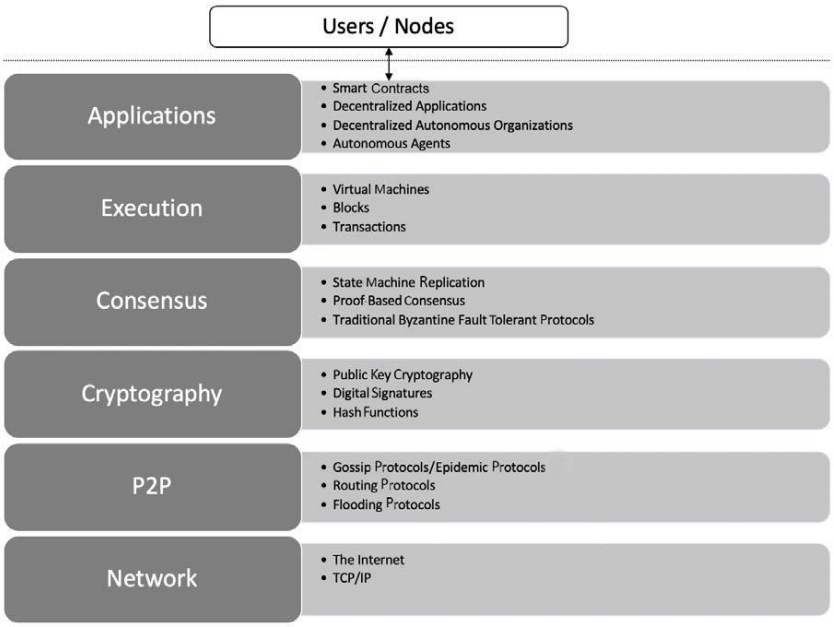


Figure 1: Blockchain Architecture and Types

3. Hyperledger Fabric Components

PEER:
Node that maintains ledger copy and executes chaincode. Validates transactions and stores blockchain records. Each organization typically operates one or more peers.

ORDERER:
Creates blocks from endorsed transactions. Ensures deterministic ordering—all peers receive transactions in identical sequence. Prevents conflicts and enables consistency.

CHANNEL:
Private communication path between subset of organizations. Transactions on a channel are visible only to channel members. Multiple channels enable parallel private networks.

LEDGER:
Combination of blockchain (transaction history) and world state (current asset values). Immutable—once recorded, transactions cannot be altered, though corrections can be added.

CHAINCODE:
Application code that implements business rules. Defines valid transactions, executes transaction logic, and updates world state. Similar to smart contracts but more flexible.

CERTIFICATE AUTHORITY (CA):
Issues digital credentials to network participants. Controls membership by validating and authenticating participants. Only holders of valid certificates can interact with the network.

4. Transaction Processing and Consensus

Hyperledger Fabric uses Byzantine Fault Tolerant (BFT) consensus. Unlike proof-of-work systems, BFT achieves agreement through multiple rounds of validation without requiring computational work.

FOUR-PHASE TRANSACTION FLOW:

PHASE 1 - ENDORSEMENT:
Client submits transaction to endorsing peers. Peers simulate transaction execution without modifying state. Each peer examines transaction and returns signed endorsement if valid, or rejection if invalid.

PHASE 2 - ORDERING:
Endorsed transaction sent to orderer service. Orderer collects endorsed transactions and sequences them deterministically. Creates block containing ordered transactions with timestamp and reference to previous block.

PHASE 3 - VALIDATION:
Orderer broadcasts block to all peers. Each peer independently validates:
- Endorsement signatures and policies
- Transaction syntax and semantic validity
- Compliance with chaincode rules

PHASE 4 - COMMIT:
If validation passes, peer commits transaction to ledger. Updates world state (current values) and appends transaction to blockchain. Ledger update is now permanent and immutable.

ADVANTAGES:

✓ No mining (energy efficient)
✓ Seconds-level finality (vs 10+ minutes for proof-of-work)
✓ No forks (deterministic ordering)
✓ 100+ transactions per second (vs 7 TPS for Bitcoin)
✓ Customizable endorsement policies

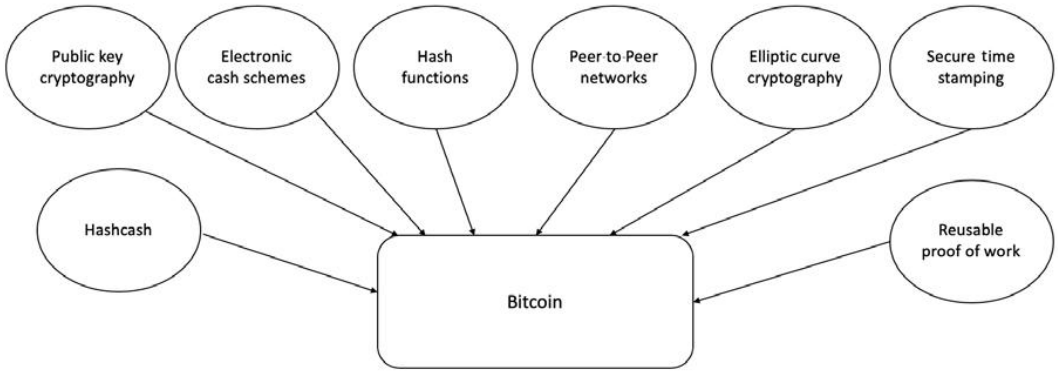


Figure 2: Consensus Mechanisms Comparison

4. Enterprise Applications

4.1 Maersk - International Shipping

BUSINESS PROBLEM:
International shipping required 10+ documents processed through different organizations with different systems. Each document required manual verification and approval through sequential steps. Total processing time: 5-7 days.

SOLUTION:
TradeLens platform built on Hyperledger Fabric creates shared ledger for shipping documents. Exporter, port authority, customs, importer, insurance, and banks access same channel. All parties view documents simultaneously and provide approvals in parallel.

BENEFITS ACHIEVED:
• Documentation time: 5-7 days reduced to 1-2 hours
• Fraud prevention: Immutable records prevent document forgery
• Dispute resolution: Digital signatures prove who authorized what, when
• Cost reduction: Fewer intermediaries, automated verification
• Scale: 400+ organizations now participate

BLOCKCHAIN ROLE:
Immutability ensures documents cannot be altered after submission. Digital signatures prove authorization. Transparency eliminates need for verification calls. Parallel processing enables instant approvals.

4.2 De Beers - Diamond Authenticity

BUSINESS PROBLEM:
Diamond fraud common: conflict diamonds sold as ethical, synthetic diamonds sold as natural, stolen diamonds mixed with legitimate inventory. Paper certificates easily forged. Buyers unable to verify authenticity.

SOLUTION:
Tracr platform records complete diamond history on Hyperledger ledger. From mining through cutting, export, retail, to customer. Each handler records:
- Unique diamond identifier
- Quality and specifications
- Handler's digital signature
- Timestamp
- Previous handler information

BENEFITS ACHIEVED:
• Authenticity verification: Customer scans product → sees complete verified history
• Counterfeit elimination: Blockchain records cannot be duplicated or forged
• Premium pricing: Verified supply command higher prices
• Industry participation: 400+ organizations across supply chain
• Regulatory compliance: Complete audit trail for certification bodies

BLOCKCHAIN ROLE:
Immutability ensures history cannot be altered. Cryptographic signatures prove identity of each handler. Transparent chain prevents substitution. Distributed copies prevent central point of failure.

5. Blockchain Technology Comparison

Feature	Bitcoin	Ethereum	Hyperledger Fabric
Architecture	Permissionless	Permissionless	Permissioned
Consensus	Proof of Work	Proof of Stake	Byzantine Fault Tolerance
Performance	7 transactions/sec	15 transactions/sec	100+ transactions/sec
Privacy	Pseudonymous (public)	Pseudonymous (public)	Private channels (controlled)
Primary Use	Digital currency	Smart contracts & DApps	Enterprise partnerships

6. Conclusion

Hyperledger Fabric represents a distinct class of blockchain technology designed specifically for enterprise applications. Unlike permissionless public blockchains, Hyperledger prioritizes:

• CONTROL: Only authorized organizations participate

• PRIVACY: Data visible only to relevant parties through channels
• PERFORMANCE: Consensus enables 100+ transactions per second
• COMPLIANCE: Regulatory requirements met through controlled access

Real-world deployments demonstrate measurable business value:
• Maersk reduced shipping documentation from days to hours
• De Beers eliminated counterfeit diamonds through immutable records
• Walmart improved food safety response from 7 days to seconds

Selection of blockchain technology depends on specific requirements:
• Require decentralization without institutions → Bitcoin
• Need programmable smart contracts → Ethereum
• Need enterprise partnership with control → Hyperledger Fabric

Understanding when and why to implement blockchain—rather than traditional databases—requires analyzing specific business problems and matching them to appropriate technology.